

Exercise 4:

Creating a baseline simulation

Policy analysis with a detailed CGE model requires two simulations. The first simulation is the baseline forecast or business-as-usual simulation. This simulation models the growth of the economy over time in the absence of the policy change under consideration. The second simulation is the policy simulation. This simulation generates a second forecast that incorporates all the exogenous features of the baseline forecast, and now includes policy-related shocks reflecting the details of the policy under consideration. The impacts of a policy are typically reported as a percentage deviation away from the baseline forecast.

In this exercise you will create the baseline forecast for the years 2006 to 2030. Due to the availability of data, you will create two different forecast closures. The first closure is created for the period 2006 to 2012. For this period we impose observed data for certain macroeconomic variables and our view on changes in productivity variables.¹ The second forecast closure is created for the period 2013 to 2030. Sometimes, in a forecast simulation we accommodate independent forecast data from reliable sources such as Treasury or other macroeconomic specialists. In this case we do not have independent forecast data. Instead, we introduce our views on selected productivity variables, labour force growth and shifts in export demand curves.

You have been provided with incomplete closure and shock files. The aim of this exercise is to complete the closure files, read in the shocks, run the simulation and interpret some results. More specifically, you will be required to complete five tasks.

You will start by creating an automatic closure. This closure is a convenient starting point to create a recursive dynamic closure. You will introduce swap statement to modify the automatic closure into a recursive dynamic closure.

You will then introduce swap statements to create the appropriate forecast closure. For the period 2006 to 2012, you will impose on the model observed data for selected macroeconomic variables. Observed data are available for variables that are naturally endogenous in IndoTERM – variables like real consumption spending. To impose data for such variables in our model, you will need a closure in which these variables are exogenous. To exogenise these variables (private consumption) it must be paired with a corresponding structural feature of the economy that is normally exogenous (such as the average propensity to consume). For the period 2013 to 2030, you will impose our views on productivity changes and labour force growth. These variables are naturally exogenous and therefore no swap statements are required.

There are a number of variables for which you have observed data. One way to specify the shock value is to type them in. However, this is too cumbersome for a large forecast simulation, so the shocks will be read from a HAR file instead. You will learn how to do this in the third part of this exercise.

Finally you will run the baseline simulation and interpret some of the results of the forecast. Remember, the forecast results will be used as the counterfactual for the policy simulation you do in the next session

¹ This closure is also called the historical closure where observed data from the statistical office are imposed on the model. Historical closures may vary depending on the availability of data. See Chapter 6 of Dixon and Rimmer (2002) for a description of the MONASH closures.

1. Getting started

Open RunDynam and choose

Zip | Restore Ingredients from ZIP Archive.

then navigate to **C:\DTERMCRS\ZIPS**, and choose **EX4.ZIP**. Unzip the ingredients to **C:\DTERMCRS\EX45**. Look at the tabbed pages Model/Data, Sim Overview, Closure/Shock, and Results.

2. Task 1: Create an automatic closure

IndoTERM is a very large dynamic regional model which includes many variables and equations. It may therefore be difficult to choose between a combination of endogenous and exogenous variables. To help setup a specific closure, we use TABmate's Tools...Closure command to create an **automatic closure**. TABmate starts from the premise that there must be the same number of endogenous variables as equations in your model. By extension, we can usefully imagine that each equation explains a particular variable. Variables not explained by any equation are deemed to be exogenous in the standard closure. In order for TABmate to know which variable is explained by a given equation, the modeller must follow a naming convention for equations. The convention is that the equation which explains variable p1, say, is named E_p1 (prefix E_). If you do not follow this convention, the Closure command will be no use to you, although the rest of TABmate will work normally.

Go to the Closure/Shock page in RunDynam. The following files should appear in this tab:

- Base closure files: COMMONXX.CLS, FCAST_1XX.CLS, FCAST_2.CLS
- Base shock files: BASEXX.BSH, FCAST_1XX.BSH, FCAST_2.BSH

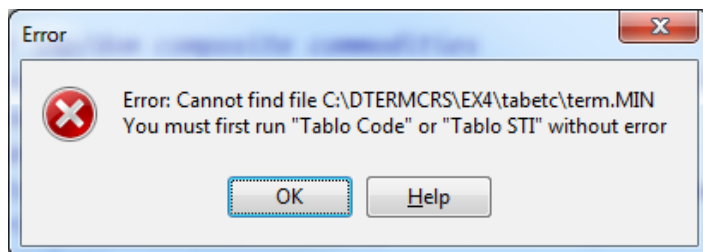
All the files with XX in their name are incomplete. You will complete these files by working through the next sections.

Part	Base Closure [B01]	Base Shocks [B01]	Policy Closure [B01]	Policy Shocks [P01]
Pattern	B01BYYYY.CLS	B01BYYYY.BSH	B01RYYYY.CLS	P01PYYYY.PSH
CMFStart	CMFSTART	N/A	none	CMFSTART
Common	COMMONXX.CLS	BASEXX.BSH	COMMONXX.CLS	none
2006	FCAST_1XX.CLS	FCAST_1XX.BSH	common	NULL.PSH
2007	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2008	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2009	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2010	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2011	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2012	FCAST_1XX.CLS	FCAST_1XX.BSH	common	none
2013	FCAST_2.CLS	FCAST_2.BSH	common	none
2014	common	common	common	none
2015	common	common	common	none
2016	common	common	common	none
2017	common	common	common	none
2018	common	common	common	none
2019	common	common	common	none

Follow Pattern As Previous Right click on cells for action Red files do not exist
Green files exist, but a constituent does not

Run Base Rerun Base Run Policy Run All Help

Open the file **COMMONXX.CLS**. You will find that the automatic closure and six swap statements. Four swap statements are blocked out with “???”. Your first task is to generate the automatic closure.² To create the automatic closure, open TERM.TAB directly in TABmate (You will find the file in C:\DTERMCRS\EX45\tabetc).³ In Tabmate click on **Tools** and then select **Closure**. If you receive the following error,



click on **OK**. In the tool bar, click on the **Tablo Code** button. Now again try to create the automatic closure. You have now created a file called TERM.CLO. Take a few minutes and review the information in this file. Scroll down to Section 4 of TERM.CLO. This section lists all the unmatched variables. If we correctly followed the naming convention in TERM.TAB, all these variables are probably exogenous. Copy and paste this list of all the exogenous variables appearing in Section 4 to COMMONXX.CLS. Copy all the comments from

```
! Automatic closure generated by TABmate Tools...Closure command
!                               Variable      Size
```

including the last comments, which is

```
Rest endogenous; ! end of TABmate automatic closure
```

and paste this list in COMMONXX.CLS in line 1. Save this closure as **COMMON.CLS**. Does this closure represent a short-run or long-run environment?

3. Task 2: Complete the swap statements

Just below the automatic closure, you will find a number of incomplete swap statements grouped into 2 sections. These swap statements change the automatic closure into a recursive dynamic closure, which is our starting point in developing an appropriate forecast closure. The swaps in the first section relates to private and public consumption and the swaps in the second section activates the dynamic mechanisms in IndoTERM. Review the swap statements and complete the 4 incomplete swap statements.

3.1 Let's look at the first swap statement. This swap endogenise total real household consumption by household and destination (xhouhtot). The aim of this swap is to link regional consumption to wage income. We do this by exogenising the household and regional-specific propensity to consume. Check the automatic closure and confirm that xhouhtot is included in the list of exogenous variables. With which variable will we swap xhouhtot? [Hint: find equation E_fhou in Term.tab]

Swap xhouhtot =

3.2. The second swap statement enforces a national rule for consumption (Swap houslack = shrBoTnom2). We make a consumption slack variable (houslack) endogenous by exogenising the ratio of the balance of trade to GDP (shrBoTnom2).

² The **Tools...Closure** command helps you to find a standard closure for your GE model. It can also be used to find logical errors in your model, or to help construct a STI file. The results of the closure analysis are contained in a text report file suffixed CLO.

³ Double-click on the desktop TABmate icon to open TABmate. Within TABmate, select **File | Open**, and navigate to the folder called C:\DTERMCRS\EX45\tabetc. Select TERM.TAB

3.3. The third swap statement endogenises a government shift variable (fgovtot) by exogenising another government shift variable (fgovtot2). Search for these variables in TERM.TAB and explain the swap statement (Swap fgovtot = fgovtot2).

3.4. The fourth swap statement activates the year-on-year capital accumulation equation by endogenising capital (xcap) and exogenising which (??) variable. [Hint: Examine the capital accumulation equation in the tab file]. Save COMMON.CLS.

Swap xcap =

3.5. The fifth swap statement switches off the investment rule used in a static simulation by endogenising an investment shift variable (finv1) and exogenising a different investment variable related to the capital accumulation mechanism. With this swap we activate the dynamic investment rule where investment is a positive function of rate of profit [Hint: examine E_finv4]. Note that swap statements four and five should be used together since they relate to capital accumulation mechanism.

Swap finv1 =

3.6. The final swap statement activates the dynamic labour market mechanism (delfwage = ??). This mechanism sets national employment as positive function of national real wage. (Hint: Search for realwage_id in the TERM.TAB and review the appropriate equation).

Swap delfwage =

After you have completed all the swap statements, save COMMON.CLS. Replace the COMMONXX.CLS file on the “Closure/Shock” page with COMMON.CLS.

The information in COMMON.CLS creates a recursive dynamic closure. Our task at hand is to modify this closure to the appropriate forecast/base case closure for the period 2006-2012.

Open **FCAST_1XX.CLS**. You will find a number of incomplete swap statements (“???”). FCAST_1XX.CLS lists six swap statements. These swaps will render exogenous the macroeconomic variables for which we have observed data available and endogenise the appropriate structural variable. Complete the statements.

3.7. The first swap statement (shrBoTnom2 = NatMacro(“RealHou”)) allows national private consumption expenditure to be determined outside the model by endogenising the ratio of BOT/GDP. Save this closure as **FCAST_1.CLS**.

3.8. The second swap statement (??? = NatMacro(“RealInv”)) makes national investments exogenous by endogenising a variable that impacts on industry and region-specific capital growth rates. Presently, investment by each regional industry is determined by rates of returns, and national investment is the sum of individual industry investments. You must find a macro structural variable that can be endogenised to accommodate forecast movements in NatMacro(“RealInv”), while allowing investment by regional industry to still be predominantly determined by movements in rates of return [Hint: search for E_gro in TERM.TAB].

swap = NatMacro(“RealInv”);

3.9. The third statement exogenously determines national government consumption spending (??? = NatMacro(“RealGov”)). Presently, real regional government consumption spending is indexed to real regional private consumption spending. You need to find a macro shift variable that allows you to exogenously determine aggregate government consumption, while still providing a role for regional private consumption in determining regional government spending. (Hint: Examine E_xgov and E_fgovtot2 in TERM.TAB and see point 3.3 above).

swap = NatMacro("RealGov") ;

3.10 The fourth swap statement exogenously determines national export volumes (natfpexp = NatMacro("ExpVol")) by endogenising a shift variable which moves the demand curve for Indonesian exports.

3.11. With the fifth swap, we wish to exogenously impose national employment data (?? = NatMacro("AggEmploy")). National employment is calculated as the share weighted sum over all regional employment. By imposing national employment data we need to endogenise a variable that allows real wage to clear. [Hint: View the equation E_delfwage].

swap = NatMacro("AggEmploy")

3.12. The final swap statement exogenously determines real GDP (swap ?? = NatMacro("RealGDP")) by endogenising the appropriate technical change variable. [Hint: Examine the equation named E_aprim].

swap = NatMacro("RealGDP") ;

Save these changes. Replace FCAST_1XX.CLS on the "Closure/Shock" page with FCAST_1.CLS, for the years 2006 to 2012.

For 2014 to 2025 we do not impose any exogenously determined forecast data in the base simulation. We therefore set the closure to "Common" which means that for each of the specified years, the valid closure is set in COMMON.CLS.

Now consider the closure used for the Base Rerun. For the dynamic term model, the policy closure is similar to the basecase closure. Thus, in the row called "Common" and column called "Policy Closure", select COMMON.CLS.

4. Task 3: Complete the shock statements

Now that we have specified the appropriate closure, we need to specify the shocks we wish to impose. Look at **BASEXX.BSH**. This file contains a list of 12 shocks. We specify this file in the Common row which means that these shocks will be introduced in every year of the baseline simulation. An alternative way to implement these shocks are to include them in every year-specific *.BSH files. However, it may be more efficient to specify the shocks in one file, eg. BASEXX.BSH and select this file in the common row. Review the shocks in BASEXX.BSH and answer the following questions.

4.1. Complete the first shock is delunity = ?. Search for delunity in TERM.TAB. What is the function of delunity? Save BASEXX.BSH as **BASE.BSH**.

delunity =

4.2. The next five shocks relates to land productivity (alnd) and the quantity of land used in the Oil Palm industry (xlnd). Shortly explain the meaning of these shocks.

4.3. The next two shocks relate to labour productivity. Explain the meaning of these shocks.

4.4. The next two shock statements increase the population and labour force growth by 1.5% and 2.5% respectively. Why do we include "uniform" in the shock statement? [Hint: search for nhou in TERM.TAB].

4.5 The next shock statement shifts the export demand for Indonesian commodities to the right by 6% per annum.

4.6. The last shock statement slightly improves the wage rate (it builds in worker's expectations of a raise).

Save this file and replace BASEXX.BSH with BASE.BSH on the “Closure/Shock” page in RunDynam.

Now open FCAST_1XX.BSH and FORECAST.HAR.⁴ The commands in these files relate to the observed macroeconomic data you wish to impose for the period 2006 to 2012. Following the syntax for the other shocks, complete the shock statement for the national private consumption in line 2.

Save FCAST_1XX.BSH as FCAST_1.BSH and carefully replace FCAST_1XX.BSH with FCAST_1.BSH on the “Closure/Shock” page. After you have changed all the files, the Closure/Shock page should look like this [no filenames should contain "XX"]:

Part	Base Closure [B01]	Base Shocks [B01]	Policy Closure [B01]	Policy Shocks [P01]
Pattern	B01BYYYY.CLS	B01BYYYY.BSH	B01RYYYY.CLS	P01PYYYY.PSH
CMFStart	CMFSTART	N/A	none	CMFSTART
Common	COMMON.CLS	BASE.BSH	COMMON.CLS	none
2006	FCAST_1.CLS	FCAST_1.BSH	common	NULL.PSH
2007	FCAST_1.CLS	FCAST_1.BSH	common	none
2008	FCAST_1.CLS	FCAST_1.BSH	common	none
2009	FCAST_1.CLS	FCAST_1.BSH	common	none
2010	FCAST_1.CLS	FCAST_1.BSH	common	none
2011	FCAST_1.CLS	FCAST_1.BSH	common	none
2012	FCAST_1.CLS	FCAST_1.BSH	common	none
2013	FCAST_2.CLS	FCAST_2.BSH	common	none
2014	common	common	common	none
2015	common	common	common	none
2016	common	common	common	none

Follow Pattern: Right click on cells for action

Red files do not exist
Green files exist, but a constituent does not

Run Base Rerun Base Run Policy Run All Help

5. Task 4: Run the simulation

You will now attempt to run the simulation for all the years of the base and base-rerun. The files COMMON.CLS, BASE.BSH, FCAST_1.CLS and FCAST_1.BSH are in the folder named “extra” in your working directory. You may either

(a) browse these files in TABMATE for the answers to the questions above, or

⁴ You will find FORECAST.HAR in your working directory for this simulation.

(b) use these files directly by moving them into your working directory and selecting them in the appropriate places on the RunDynam Closure/Shocks page.

We suggest option (a) if you have completed or almost completed the questions above, and option (b) otherwise.

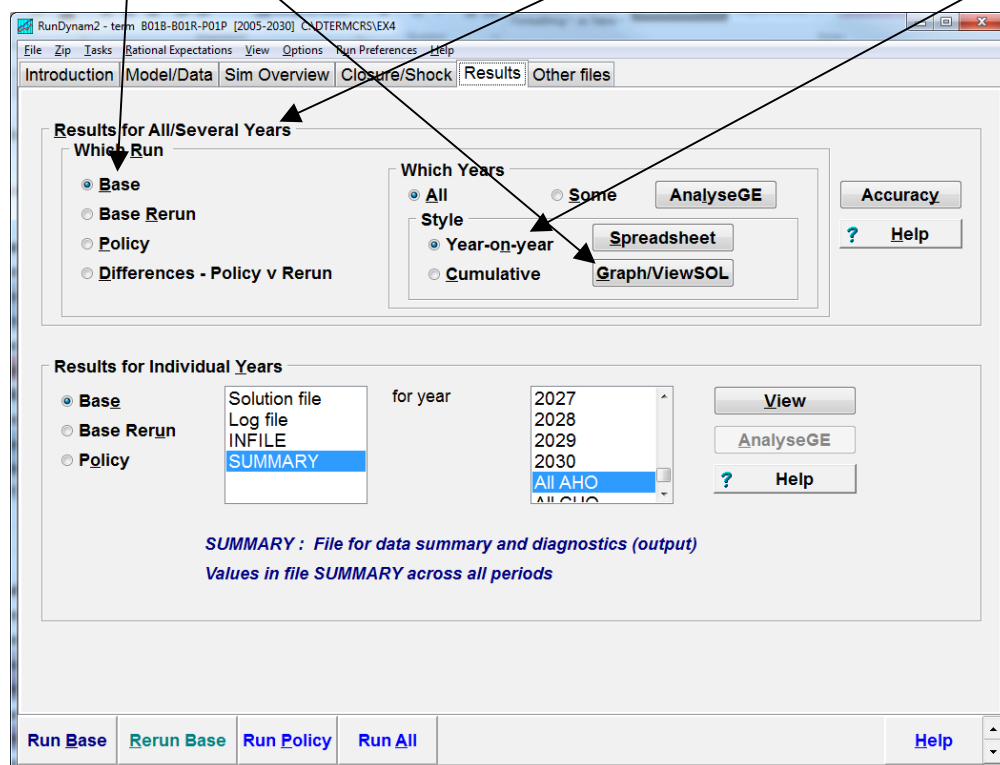
If you click the Run Base button in the bottom left hand corner, RunDynam will run the simulation for all years. You may wish to run the baseline for only a selected number of years. To do this, go to the **Task** menu and select **Run Part Base, Same Part Base Rerun**. There are many options under the **Task** menu. Familiarity with these options may save you a lot of time! In this exercise we are going to run the base and rerun for all the years. First, check that your closure and shocks are ok. Under the **Task** menu select **Check all Base-case closures and shocks**. RunDynam will check all the closure and shock files. If you do not get the message “Check of all Base-case closures and shocks completed”, return to option (b) in the previous paragraph.

To run the simulation, click on the **Task** menu and select **Do Base and Base Rerun**. Click **OK**.

6. Task 5: Interpret the results

6.1 Base Run

Has the simulation run as you expected? Check the Base Run. The top half of the Results page in RunDynam gives options for viewing Results for all or several years. View the Year-on-year results for the Base, in ViewSQL.



In ViewSol look for the variable called NatMacro. Check total real private consumption (RealHou). Is it exogenous (red) or endogenous (black)? How much does it increase in 2006? Check FORECAST.HAR. Is this as you expected?

The first line in ViewSol contains a variable called Macros. When you click on Macros, you will find results for scalar variables. Check invslack. Is it exogenous (red) or endogenous (black)? What is the percentage change for 2006?

6.2 Base Rerun

Now check the Base Rerun. Close ViewSol and return to the Results page in RunDynam. Select Year-on-year results for the Base Rerun.

Go to the NatMAcro variable and check total real private consumption. Is it exogenous or endogenous? How much does it increase in 2006?

Check invslack. Is it exogenous or endogenous? How much does it change in 2006?

6.3 Comparing the Base and Base Rerun

Now check the Base and Base Rerun side by side. In ViewSol, select **File | Close All** and then open the solution files for 2006 by selecting **File | Open**. You will find the solution files in the working directory. The base is called *b01b-2006.sl4* and the Base Rerun is called *b01b-b01r-2006.sl4*. The two solution files appear side by side.

Look at the NatMacro variable. What do you notice about the variables which are exogenous in the Base and endogenous in the Base Rerun?

Look at the Macros screen. What do you notice about the variables which are exogenous in Base Rerun and endogenous in Base?

Close the solution files in ViewSOL.

6.4 Review results for all the years

Open the Base Rerun results in cumulative style. Open the results for national industry output (natxtot). Draw a graph, using the Chater function, of industry output by clicking on “natxtot” in the top left hand corner. Which four industries are forecast to expand the most by 2030? Which four industries are forecast to contract or expand the least? Write some short notes explaining:

- the favourable output outcomes for the top three industries [Hint: compare the forecasts for aggregate exports and aggregate investment with the forecast for real GDP. Examine the household expenditure elasticities. Examine the forecast shocks to productivity]; and
- the unfavourable output outcomes for the bottom three industries. [Hint: compare the forecasts for aggregate imports and real GDP, and examine the household expenditure elasticities and productivity shocks]

Open the results for real GDP expenditure by region (xgdpexp). Draw a graph, using the Chater function. Which regions performs the best and worst?